

Piezoelectrically induced electron detrapping model for mechanoluminescent semiconductors: A unified theoretical and experimental perspective

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ABSTRACT

Mechanoluminescence (ML) in semiconductors originates from the coupled interaction between mechanical stress, internal polarization fields, and trapped charge carriers. Rather than proposing a new mechanism, this perspective synthesizes and critically examines the existing theoretical and experimental studies that attribute stress-induced light emission to piezoelectrically assisted electron detrapping in semiconducting and phosphor materials. Emphasis is placed on frameworks developed for systems such as ZnS:Mn and ZnO, where mechanically generated piezoelectric potentials modulate trap depths, facilitate carrier release, and enable radiative recombination. By revisiting established formulations and extending their discussion to include non-uniform field distributions, multi-level trap states, and realistic defect landscapes, this perspective highlights how piezoelectric field effects can coexist with, complement, or dominate over thermal and triboelectric contributions depending on material chemistry, defect structure, temperature, and loading conditions. Comparative analysis underscores that piezoelectric assisted detrapping is particularly effective under dynamic stress and low-temperature regimes, while thermally activated processes remain robust and reliable in other contexts. Together, these insights provide a balanced interpretative framework for understanding ML across material classes and offer guiding principles for the rational design of self-powered mechanoluminescent and piezophotonic systems for stress sensing, structural health monitoring, and energy-autonomous optoelectronic applications and to identify key directions for future research.

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I. INTRODUCTION

Mechanoluminescence (ML), the emission of light induced by mechanical stimulation, has attracted sustained scientific interest due to its relevance in stress sensing, energy harvesting, and self-powered optoelectronic systems.^{1–3} Over the past decades, several mechanisms have been advanced to explain ML phenomena, among which piezoelectric field-assisted electron detrapping has emerged as a particularly influential framework linking mechanical deformation with charge carrier excitation and radiative recombination.^{4,5} In widely studied semiconducting phosphors such as ZnS:Mn and ZnO, applied mechanical stress generates internal piezoelectric potentials that locally modulate trap depths, thereby

facilitating the release of charge carriers and subsequent light emission.^{6,7} Early interpretations of mechanoluminescence relied predominantly on triboelectric charging or thermally assisted detrapping processes; however, these approaches are often insufficient to explain the fast temporal response, reversibility, and spatial localization of emission observed in non-centrosymmetric and piezoelectric materials.⁸ Recent advances in time-resolved photoluminescence spectroscopy, Kelvin probe force microscopy, and nanoscale electromechanical characterization have provided compelling evidence that stress-induced piezoelectric fields can dynamically alter band structures and significantly enhance detrapping probabilities under mechanical loading.^{9,10} These findings have contributed to a growing consensus that mechanoluminescence in

many semiconductors is best interpreted as a field-assisted emission process rather than a purely thermal or contact-driven effect.

Despite this progress, existing descriptions often remain fragmented, with many models assuming uniform stress fields, simplified trap distributions, or idealized crystal structures. In practice, real ML materials exhibit complex defect landscapes, including grain boundaries, dislocations, and multiple trap levels, all of which strongly influence carrier dynamics and emission efficiency.¹¹ Moreover, systematic experimental validation across emerging material classes such as hybrid semiconductors and perovskite-derived systems remains limited, leaving open questions regarding the generality and boundaries of field-assisted ML frameworks.¹²

In this context, this perspective highlights and synthesizes existing theoretical and experimental models of piezoelectrically assisted electron detrapping, drawing particular attention to their shared physical principles and practical limitations. Rather than proposing a new mechanism, this article advocates a unifying viewpoint that integrates piezoelectric potential generation, trap-depth modulation, and defect-assisted recombination to interpret stress-driven luminescence across diverse material systems. Comparative discussions with triboelectric and thermally activated detrapping frameworks are included to clarify the conditions under which each mechanism dominates and to emphasize their complementary roles. From an application standpoint, mechanoluminescent materials offer an intrinsic and energy-autonomous route for converting mechanical deformation directly into optical signals, enabling real-time stress and strain visualization without external power supplies or complex circuitry. Such capabilities make ML based platforms particularly attractive for nondestructive monitoring in optically opaque structures, confined geometries, and biologically complex environments.^{13–15} By consolidating current understanding and identifying open challenges, this perspective aims to provide guidance for the rational design of next generation mechanoluminescent and piezophotonic systems. To ensure a comprehensive and up to date perspective, this review incorporates recent experimental and theoretical advances in mechanoluminescent materials, including perovskite-based systems, rare-earth-doped oxide phosphors, and modern characterization techniques, supported by newly reported high resolution experimental and mechanism based figures from the recent literature.

II. ELECTRON TRAPPING AND DETRAPPING IN PIEZOELECTRIC MATERIALS

In piezoelectric materials, electron trapping (ET) occurs when charge carriers are captured by defect states within the crystal lattice, while detrapping involves their release under external stimuli, such as mechanical stress. When stress is applied, the resulting piezoelectric field reduces the effective trap depth, facilitating electron detrapping. The release of trapped electrons into the conduction band enables their subsequent recombination with holes or luminescent centers, thereby giving rise to emission processes such as ML. Figure 1 schematically illustrates the One-Trap–One-Recombination (OTOR) model, which was originally developed within the theoretical framework of thermoluminescence (TL) to describe thermally activated carrier release from defect states. Although first formulated for TL phenomena, the OTOR

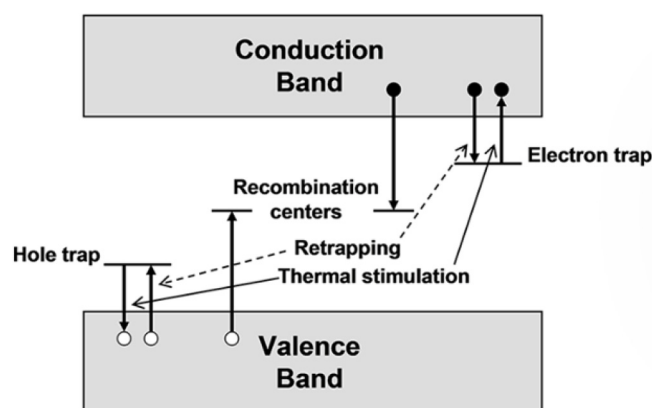


FIG. 1. Basic mechanism of charge trapping and detrapping for electron and hole traps. Reproduced with permission from Varney *et al.*, J. Appl. Phys. 111, 063520 (2012). Copyright 2012 American Institute of Physics.¹⁶

concept has since been adapted to mechanoluminescent systems, where mechanical stress rather than thermal energy serves as the stimulus for carrier detrapping. In this perspective, the OTOR framework is employed as a conceptual foundation to discuss field-assisted detrapping under piezoelectric excitation.⁷ The detrapping rate depends on factors such as the piezoelectric constant, trap depth, and applied pressure. In materials with shallow traps or high piezoelectric activity, detrapping occurs more readily, enhancing luminescence or conductivity. It is important to clarify that the electron detrapping framework discussed in this perspective does not assume external excitation (such as UV irradiation) to be a universal prerequisite for ML. Early trap-assisted descriptions of ML, many of which were adapted from classical TL theory, implicitly considered pre-filled traps populated through optical or thermal charging. However, extensive experimental evidence demonstrates that a large class of ZnS-based phosphors exhibits intense ML without any deliberate pre-irradiation, commonly described as self-powered or self-charging ML behavior.

In these materials, trap occupation can arise intrinsically from defect-mediated carrier generation, internal band bending at grain boundaries, or built-in electric fields associated with piezoelectric polarization. Under mechanical deformation, stress, therefore, plays a dual role: it not only generates internal piezoelectric fields that lower effective trap depths and promote electron detrapping but can also facilitate continuous trap refilling through stress-induced carrier separation and interfacial charge redistribution. Consequently, the piezoelectrically assisted detrapping framework should be regarded as a generalized description that accommodates both pre-charged and self-powered ML systems, with the dominant trap population pathway governed by material composition, defect landscape, microstructural interfaces, and mechanical loading conditions.

This distinction is essential for correctly interpreting ML in ZnS-based and related semiconductors and for avoiding the misconception that external excitation is an inherent requirement of the detrapping process.

The interplay between trapping and detrapping is critical for applications in stress sensing, energy harvesting, and optoelectronic devices, where controlled charge carrier dynamics are essential. Understanding these mechanisms enables the design of advanced piezoelectric materials with tailored trap states for improved performance.²

A. Significance of piezoelectric fields in charge carrier dynamics

The significance of piezoelectric fields in charge carrier dynamics lies in their ability to influence electron behavior in non-centrosymmetric materials like ZnS. When mechanical stress is applied, piezoelectric fields are generated, which can reduce the effective depth of electron traps near luminescent centers such as Mn^{2+} ions. This reduction in trap depth facilitates the detrapping of electrons into the conduction band, where they recombine with holes, releasing energy non-radiatively. This energy can then excite luminescent centers, leading to light emission. The process is reversible and reproducible, as the piezoelectric fields also arise during stress release, ensuring consistent charge carrier dynamics. The strength of these fields determines the efficiency of electron detrapping and subsequent recombination, making them crucial for understanding and optimizing light emission in piezoelectric materials. This mechanism highlights the interplay between mechanical energy, piezoelectric effects, and charge carrier dynamics, offering insights for designing advanced optoelectronic and stress-sensing applications.²

III. THEORETICAL FRAMEWORK OF THE MODEL

ZnO is first considered a model piezoelectric semiconductor to demonstrate the generation of stress-induced piezoelectric potential and its role in modulating charge carrier behavior, owing to its well-known non-centrosymmetric crystal structure and pronounced piezoelectric response. For the experimental verification of mechanoluminescence, ZnS:Mn^{2+} is employed due to its extensively reported elastico-mechanoluminescent characteristics, clearly defined trap states, and reproducible response under dynamic mechanical loading. Despite differences in crystal chemistry, both material systems operate under the same piezoelectrically assisted electron detrapping mechanism, thereby maintaining consistency between the theoretical framework and experimental observations.

A. Piezoelectric potential generation

This perspective draws on established piezoelectric theory, in which electromechanical coupling between applied stress and internal electric fields in non-centrosymmetric semiconductors such as GaN, AlGaN, ZnO, and ZnS-based phosphors governs the generation of stress-induced piezoelectric potentials that modulate charge carrier dynamics. When an external mechanical stress or electric field is applied, the piezoelectric effect generates an internal polarization, leading to the creation of a piezoelectric potential. This potential arises due to the displacement of ions in the crystal lattice, which produces a localized electric field. In materials like GaN-based HEMTs, this effect is particularly significant because of

their inherent piezoelectric properties. Figure 2(a) shows that the piezoelectric coefficient d_{33} signifies that the electric potential is generated along direction 3 when mechanical stress is applied in the same direction 3. Mechanical-stress-induced deformation causes a relative shift between the centers of positive and negative charges in the crystal structure of the material, leading to the formation of electric dipoles or polarization as shown in Fig. 2(b). This aligned and distorted dipole state generates an electric potential, resulting in a flow of charge. This phenomenon arises from the non-centro-symmetric structure characteristic of piezoelectric crystals, where applied stress disrupts the charge balance to produce surface charges.¹

The generated potential can influence charge carrier dynamics, such as electron trapping or detrapping, by altering the energy barriers within the material. For instance, under reverse-bias conditions, the piezoelectric field can modify the stress distribution in the barrier layer, affecting the critical voltage for degradation. The previous model accounts for the relationship between the applied electric field, mechanical stress, and the resulting piezoelectric response, providing insights into device behavior under varying conditions. This framework is essential for understanding reliability issues in piezoelectric materials and optimizing their performance in applications like power electronics or sensors. The interplay between stress, electric fields, and charge carriers forms the basis for predicting and mitigating device degradation.⁸

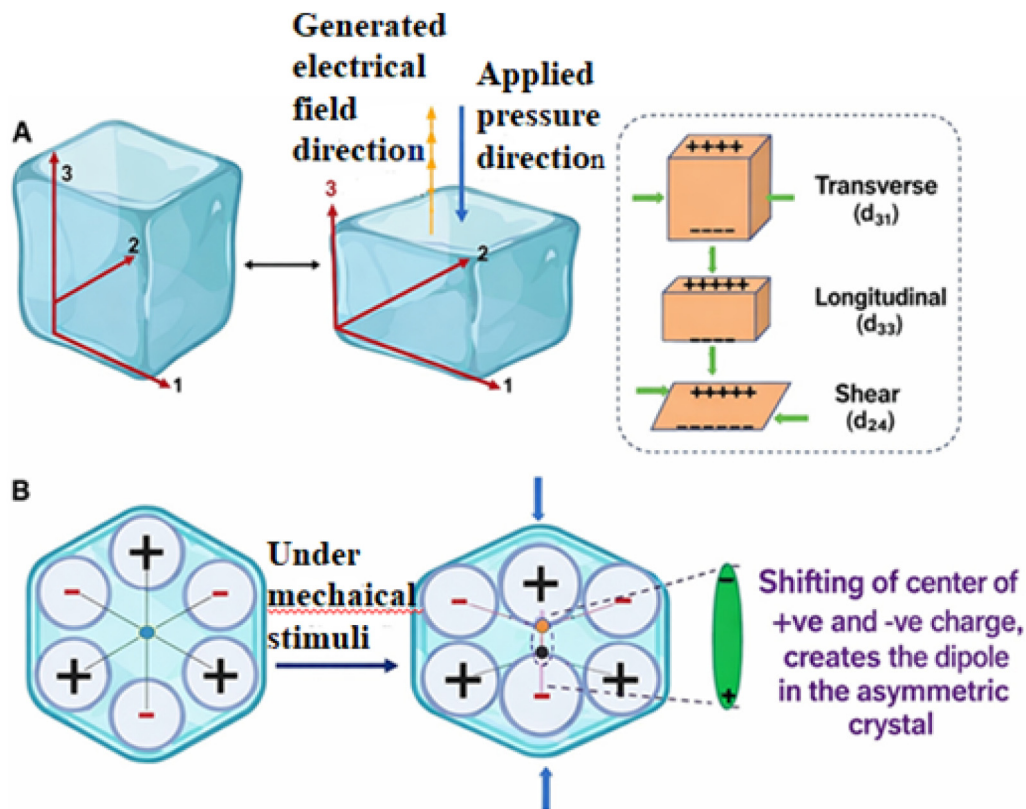
B. Electron detrapping mechanisms

1. Poole-Frenkel effect

Field-assisted detrapping, particularly through the Poole-Frenkel effect, describes how trapped electrons escape from defect states in a material under the influence of an external electric field. In this mechanism, the applied electric field lowers the potential barrier of electron traps, making it easier for carriers to overcome the confinement energy through thermal activation. The Poole-Frenkel effect is characterized by a logarithmic dependence of the detrapping rate on the electric field strength, where the effective trap depth is reduced proportionally to the square root of the field. This process is common in insulating and semiconducting materials, especially those with deep-level defects. Unlike direct tunneling, which is temperature-independent, Poole-Frenkel emission is thermally activated, meaning that higher temperatures enhance detrapping efficiency. Figure 3 shows the energy band diagram illustrating the Poole-Frenkel effect and the trap-assisted tunneling process. The phenomenon plays a critical role in charge transport, leakage currents, and reliability issues in electronic devices. By understanding and controlling field-assisted detrapping, material properties can be optimized for applications such as resistive switching, memory devices, and sensors, where precise charge carrier dynamics are essential. The Poole-Frenkel model provides a foundational framework for analyzing trap-dominated conduction and degradation mechanisms in solid-state systems.⁹

2. Tunneling

In certain materials, electron detrapping can occur through tunneling, a quantum mechanical process where electrons



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FIG. 2. (A) The d_{33} piezoelectric coefficient represents the generation of electrical potential along axis 3 when mechanical stress is applied along the same axis (direction 3). (B) The underlying mechanism of piezoelectric potential formation in crystalline materials occurs due to the asymmetric charge distribution within the unit cell. Reproduced with permission from Ghosh *et al.*, *J. Funct. Biomater.* **14**(1), 8–36 (2022). Copyright 2022 Author(s), licensed under a Creative Commons Attribution (CC BY) license.¹

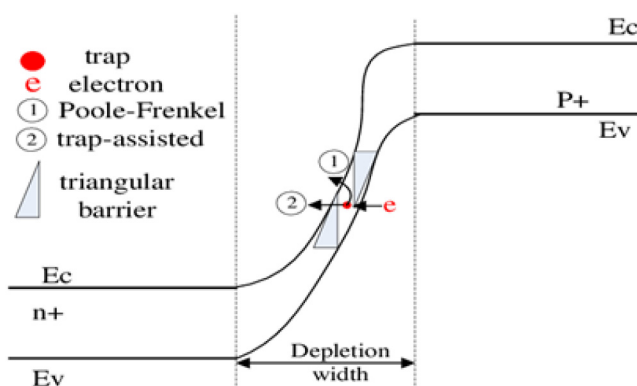


FIG. 3. Energy band representation highlighting Poole-Frenkel barrier lowering and carrier transport through trap-assisted tunneling. Reproduced with permission from Wang *et al.*, *J. Phys.: Conf. Ser.* **1802**(3), 032083 (2021), licensed under a Creative Commons Attribution (CC BY 3.0) license.¹⁰

escape from traps without requiring thermal energy. When a local piezoelectric field is induced by mechanical stress, it can cause band bending near defect centers or activator ions. This bending reduces the effective barrier height, enabling trapped electrons to tunnel directly into the conduction band. The tunneling probability depends on the electric field strength and the trap depth, with higher fields or shallower traps increasing the likelihood of tunneling. Once in the conduction band, these electrons may recombine at luminescent centers, contributing to light emission. This mechanism is particularly relevant in materials with high piezoelectric constants or localized distortions, where strong electric fields facilitate tunneling. Unlike thermal detrapping, tunneling is independent of temperature, making it efficient even at low temperatures. The process is often observed in persistent luminescent materials, where mechanical deformation triggers rapid electron release through tunneling, followed by radiative recombination. This tunneling-based detrapping is a key factor in the mechanoluminescent response of certain crystals and thin films.¹¹

3. Critical thresholds for trap ionization

In the piezoelectrically induced electron detrapping model, critical thresholds determine when trapped electrons are released under mechanical stress. When stress is applied, non-centrosymmetric materials generate localized piezoelectric fields near defect centers, which reduce the effective trap depths. However, detrapping only occurs when the piezoelectric field exceeds a threshold value, F_{th} , necessary to sufficiently lower the potential barrier. Below this threshold, traps remain stable, and no significant electron release happens. The detrapping process follows an exponential distribution, where deeper traps require stronger fields for ionization. As stress increases, the piezoelectric field grows, enhancing the detrapping rate until most electrons are released into the conduction band. This mechanism explains why mechanoluminescence is only observed above certain stress levels: sub-threshold stresses fail to generate the required field strength. Materials with higher piezoelectric coefficients or greater defect densities exhibit lower thresholds due to stronger local field effects. The critical threshold concept is essential for optimizing mechanoluminescent materials, as it governs their stress sensitivity and emission efficiency. By controlling trap depths and piezoelectric response, materials can be tailored for specific applications, such as stress sensors, where precise threshold tuning ensures reliable operation under desired mechanical conditions.¹²

C. Mathematical formulations

In the piezoelectrically induced electron detrapping model, the relationship between piezoelectric charge density and detrapping rate is governed by key equations.

When mechanical stress is applied, the piezoelectric field F is generated, which reduces the effective trap depth for electrons. The effective trap depth under an applied piezoelectric field is given by the following equation:

$$E_{t, eff} = E_t - \alpha F, \quad (1)$$

where $E_{t, eff}$ represents the effective trap depth and α denotes the trap-depth reduction per unit electric field. This means the electron needs less energy to escape the trap under a higher field F . The detrapping probability is given by the following equation:

$$g(F) = \exp(-E_{t, eff}/kT). \quad (2)$$

Here, the detrapping probability $g(F)$ follows an Arrhenius type thermal activation form with a negative exponential dependence, consistent with established thermoluminescence and field-assisted emission theory.

Let $Z = 1/kT$,

$$g(F) \propto \exp[-Z(E_t - \alpha F)]. \quad (3)$$

Since ZE_t is a constant, we focus only on the field dependent term,

$$g(F) \propto \exp(Z\alpha F). \quad (4)$$

This indicates that as the applied field increases, the effective trap barrier is reduced and the detrapping probability rises exponentially. The parameters Z and α quantify the sensitivity of the traps to the field-induced lowering of the activation energy.

The detrapping rate g of electrons is given by the following equation:

$$g = N_0 Z \alpha \dot{F} \exp(-Z\alpha Ft). \quad (5)$$

Here, N_0 is the initial trapped electron density, Z is the trap distribution coefficient, and $\dot{F}$ is the rate of change of the piezoelectric field. The term $\exp(-Z\alpha Ft)$ accounts for the exponential decrease in detrappable traps over time.

The piezoelectric charge density Q is linked to the applied stress P via the piezoelectric constant d_0 , given by the following equation:

$$Q = d_0 P. \quad (6)$$

For a correlating factor B , the piezoelectric field becomes $F = B d_0 P$. Substituting this into the detrapping rate equation shows that higher stress P or faster pressing rates $\dot{P}$ increase the detrapping rate. The resulting electron density in the conduction band Δn is given by the following equation:

$$\Delta n = g\tau = N_0 Z \alpha \tau B d_0 \dot{P} \exp(-Z\alpha B d_0 Pt), \quad (7)$$

where τ represents the electron lifetime. It provides a useful framework for understanding transient carrier generation in piezoelectric materials under dynamic stress. This formulation captures the transient nature of carrier generation in a piezoelectric material subject to time-varying mechanical stress. The term $B d_0 \dot{P}$ accounts for the rate of stress-induced polarization, which directly influences the generation rate of free carriers via the piezoelectric effect.

The exponential decay component $\exp(-Z\alpha B d_0 Pt)$ is an empirical approximation for time-dependent loss mechanisms and is introduced to represent carrier recombination and field attenuation effects over time.

These equations highlight how piezoelectric charge density directly influences detrapping dynamics, enabling quantitative prediction of ML intensity under varying stress conditions.¹⁷

The above formulation assumes a spatially uniform piezoelectric field and homogeneously distributed traps. In practical materials, however, stress gradients, grain boundaries, and defect clusters generate localized non-uniform fields that alter detrapping probabilities. To capture these effects, the previous model can be extended by introducing a position-dependent field $F(x, t)$ and a multi-level trap distribution $N(E, t)$, yielding more realistic predictions for heterogeneous microstructures.

D. Time-dependent vs steady-state models

The piezoelectrically induced electron detrapping model can be analyzed through both time-dependent and steady-state formulations. In the time-dependent model, the local piezoelectric field generated by mechanical stress reduces the trap depth of electrons, enabling their detrapping. The detrapped electrons move in the

conduction band and are captured by activators like Eu^{2+} ions, leading to light emission upon de-excitation. The ML intensity $I(t)$ is proportional to the product of the applied pressure $P(t)$ and its rate of change $\dot{P}(t)$, expressed as

$$I(t) \propto P(t)\dot{P}(t). \quad (8)$$

The peak ML intensity occurs at a specific time t_m , derived from the pressure rise dynamics, and the decay profile reflects the recombination kinetics of charge carriers. In contrast, steady-state models assume a constant piezoelectric field, where the detrapping rate balances recombination, resulting in a stable ML output. These formulations highlight the interplay between mechanical energy input and electronic transitions, providing a framework to predict ML behavior under varying stress conditions. The previous models emphasize the role of piezoelectric fields in governing electron dynamics, offering insights for designing ML based sensors and light sources.¹⁸

IV. MATERIAL SYSTEMS AND DEFECT ENGINEERING

A. Key piezoelectric materials

1. Inorganic semiconductors

The piezoelectrically induced electron detrapping model is particularly relevant in inorganic semiconductors such as ZnO, GaN, and perovskites, where mechanical stress generates piezoelectric fields that influence charge carrier dynamics. In these materials, localized piezoelectric fields near dopant sites or defects can reduce trap depths, facilitating electron detrapping. For instance, in Mn doped ZnS nanocrystals, mechanical deformation creates piezoelectric fields that release trapped electrons into the conduction band, where they recombine with activators to produce luminescence. Similarly, in ZnO and GaN, stress-induced piezoelectric fields can modulate carrier transport and recombination, enhancing

optoelectronic responses. Perovskites, with their flexible lattice structures, exhibit strong piezoelectric effects under strain, which can be harnessed for efficient electron detrapping and light emission. The previous model suggests that the ML intensity in such materials depends on the magnitude of the piezoelectric field, which scales with applied stress. Figure 4(a) shows the atomic model of ZnO, and Fig. 4(b) shows the piezoelectric properties and the difference of piezopotential in tension and compression. This mechanism is distinct from dislocation-based models, as it emphasizes the role of piezoelectricity rather than dislocation motion. The behavior aligns with observations where the ML intensity correlates quadratically with stress, supporting the dominance of piezoelectric effects in these systems. Understanding this process facilitates the design of stress-sensitive optoelectronic devices and sensors based on inorganic semiconductors.²⁰

2. Trap-rich phosphors

The piezoelectrically induced electron detrapping model is particularly relevant for trap-rich phosphors like $\text{SrAl}_2\text{O}_4:\text{Eu}^{2+}$ and $\text{ZnS}:\text{Mn}$, where mechanical stress generates localized piezoelectric fields that facilitate charge carrier dynamics. In these materials, dopants such as Eu^{2+} or Mn create defect states that act as electron traps. When stress is applied, the resulting piezoelectric field reduces the effective trap depth, enabling trapped electrons to be released into the conduction band. For instance, in $\text{SrAl}_2\text{O}_4:\text{Eu}^{2+}$, the piezoelectric field near Eu^{2+} ions detraps electrons from oxygen vacancies, which then recombine with luminescent centers to produce light. Similarly, in $\text{ZnS}:\text{Mn}$, stress-induced piezoelectric fields near Mn sites release trapped electrons, leading to radiative recombination and ML.²¹ To provide a comparative and comprehensive understanding of the piezoelectrically driven mechanoluminescence (ML) process, Fig. 5 has been expanded to include parallel schematics for both $\text{SrAl}_2\text{O}_4:\text{Eu}^{2+}, \text{Dy}^{3+}$, and $\text{ZnS}:\text{Mn}^{2+}$ systems. While $\text{SrAl}_2\text{O}_4:\text{Eu}^{2+}$, Dy^{3+} represents a trap-rich oxide

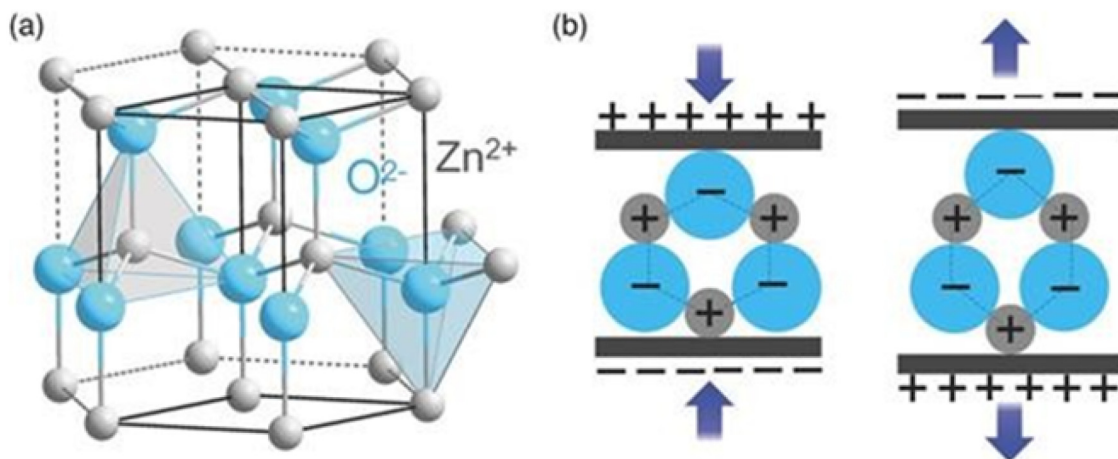


FIG. 4. (a) Atomic structure of ZnO and (b) piezoelectric response of ZnO, highlighting the difference in piezopotential generated under tensile and compressive deformation. Reproduced with permission from Fan *et al.*, Adv. Mater. **28**(22), 4283–4305. Copyright 2016 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim.¹⁹

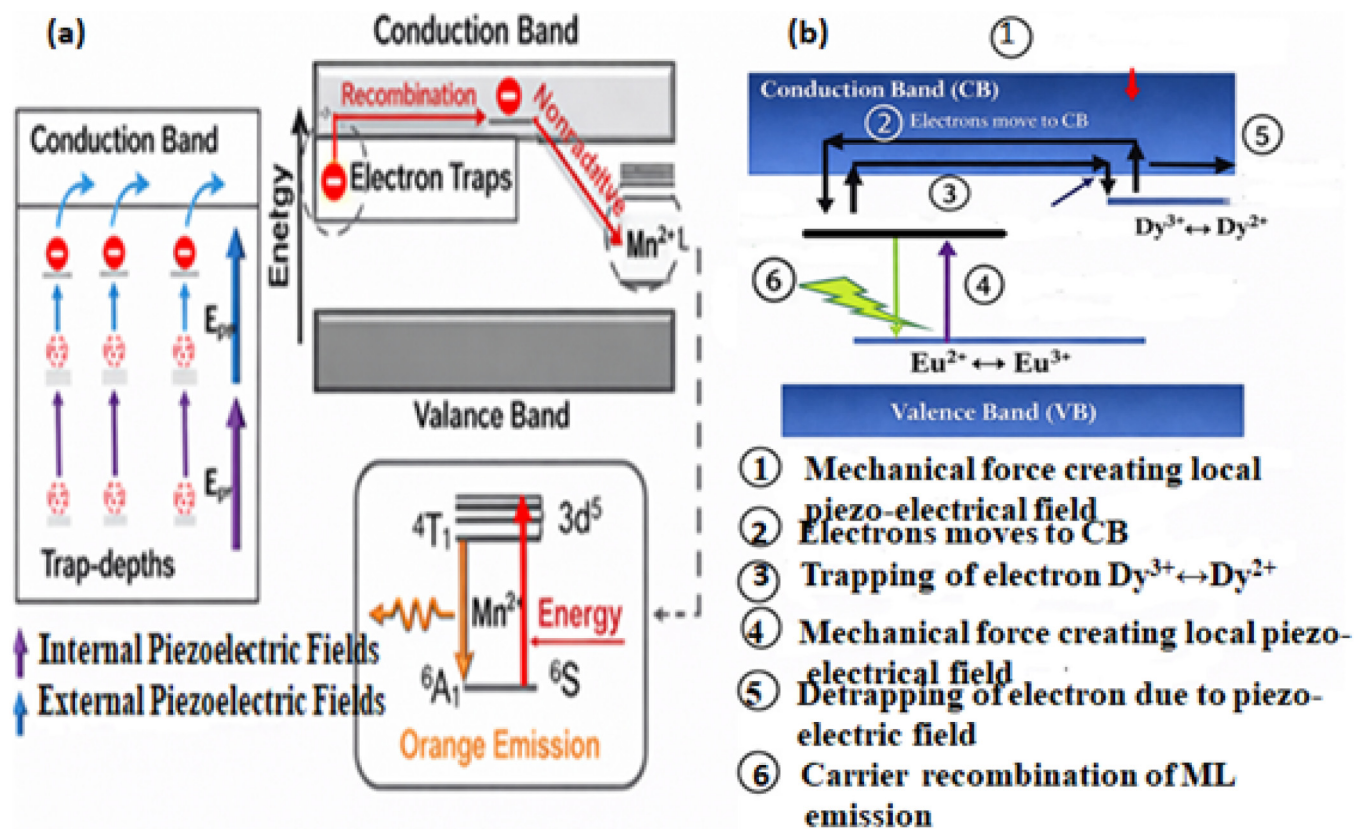


FIG. 5. Comparative schematics of piezoelectrically driven mechanoluminescence mechanisms in (a) ZnS:Mn²⁺ and (b) SrAl₂O₄:Eu²⁺, Dy³⁺. In both systems, mechanical stress generates localized piezoelectric fields that reduce trap depths and promote electron detrapping into the conduction band, followed by radiative recombination at activator centers. While ZnS:Mn²⁺ involves sulfur vacancy related traps and Mn²⁺ emission, SrAl₂O₄:Eu²⁺, Dy³⁺ exhibits rare-earth-assisted trapping and recombination pathways. Despite differences in the host lattice and defect chemistry, the underlying piezoelectrically assisted electron detrapping mechanism remains common to both materials.

phosphor exhibiting long-lasting charge storage and delayed recombination, ZnS:Mn²⁺ serves as a classical elasto-mechanoluminescent semiconductor with relatively shallow traps and prompt emission under dynamic stress. In both materials, mechanical deformation generates localized piezoelectric fields that reduce effective trap depths, enabling electron detrapping into the conduction band followed by radiative recombination at the respective activator centers (Eu²⁺ or ZnS:Mn²⁺). Despite differences in host lattice chemistry, defect nature, and emission kinetics, the fundamental mechanism of piezoelectrically assisted electron detrapping remains common to both systems, underscoring the general applicability of previously reported theoretical frameworks. The ML intensity in these systems often scales quadratically with applied stress, reflecting the dominance of piezoelectric effects over dislocation-based mechanisms. The detrapped electrons may either recombine directly with activators or migrate through the conduction band before emission. Existing studies indicate that, in trap-rich phosphors, piezoelectric effects relevant to mechanoluminescence are frequently localized near dopant induced lattice

distortions rather than arising from uniform bulk polarization. Such insights are valuable for designing stress-sensitive luminescent materials for sensors and optoelectronic applications.²²

Group III-N semiconductors such as GaN, AlN, and InGaN exhibit strong piezoelectric coupling and have been widely used in piezophototronic devices, where mechanical deformation modulates carrier injection in LEDs or photodetectors. Incorporating such systems into the detrapping framework could further connect macroscopic piezoelectric potentials with nanoscale emission control.

B. Role of defects and dopants

Defects and dopants play a critical role in tailoring trap depths and densities for controlled electron detrapping in piezoluminescent materials. In materials like SrAl₂O₄:Eu²⁺ and ZnS:Mn, dopants such as Eu²⁺ or Mn introduce localized defect states that act as electron traps. These traps can be strategically modified to optimize detrapping under mechanical stress. For instance, in

$\text{SrAl}_2\text{O}_4\text{:Eu}^{2+}$, oxygen vacancies and Eu^{2+} ions form shallow traps, while codoping with Dy^{3+} creates deeper traps, enabling persistent luminescence. The trap depths influence the efficiency of piezoelectrically induced detrapping, as shallow traps require lower piezoelectric fields for electron release compared to deeper ones.

In ZnS:Mn , Mn^{2+} dopants distort the local lattice, creating piezoelectric-active regions where stress-induced fields reduce trap depths, facilitating electron detrapping. The density of these traps can be tuned by adjusting dopant concentrations, with higher densities leading to stronger ML but potentially causing quenching at excessive levels. Centrosymmetric materials like $\text{ZrO}_2\text{:Ti}$ exhibit ML only when doped, as Ti ions introduce localized piezoelectricity near defects, enabling detrapping despite the bulk non-piezoelectric structure.

Thus, defects and dopants serve as design tools to control trap properties, ensuring efficient detrapping under stress. This approach allows the customization of ML intensity and duration, critical for applications like stress sensors or light sources. The interplay between trap depth, density, and local piezoelectricity underscores the importance of defect engineering in piezoluminescent materials.²³

C. Microstructural influences

The piezoelectrically induced electron detrapping model is influenced by microstructural features such as grain boundaries, interfaces, and strain gradients within the material. These structural inhomogeneities can alter the local piezoelectric fields, affecting the efficiency of electron detrapping and subsequent luminescence. Grain boundaries and interfaces often act as sites for localized defects or trap centers, where the piezoelectric field may be enhanced due to structural discontinuities. Strain gradients, arising from mechanical deformation, can further modify the distribution of piezoelectric charges, leading to non-uniform detrapping rates across the material. Such microstructural effects are critical in determining the overall mechanoluminescent response, as they influence the mobility of detrapped electrons and their recombination pathways. For instance, regions with high strain gradients may exhibit stronger luminescence due to intensified piezoelectric fields, while grain boundaries could either facilitate or hinder electron transport depending on their electronic properties. Understanding these microstructural influences is essential for optimizing material design to achieve consistent and efficient mechanoluminescence in applications like stress sensing or mechano-optical devices. By tailoring the microstructure, it is possible to enhance the performance of piezoelectrically active materials under mechanical stimuli.²⁴

V. MATERIAL-SPECIFIC DETRAPPING MECHANISMS

A. Oxide-based piezoelectrics (e.g., ZnO , $\text{SrAl}_2\text{O}_4\text{:Eu}$)

In oxide-based piezoelectric materials such as ZnO and $\text{SrAl}_2\text{O}_4\text{:Eu}$, intrinsic defects particularly oxygen vacancies play a crucial role in enabling piezoelectrically assisted carrier detrapping. These defects introduce localized energy states within the bandgap that act as trapping centers for charge carriers. When mechanical stress is applied, the resulting piezoelectric polarization generates an internal electric field that alters the local potential distribution

and reduces the effective trap depth. This facilitates the release of trapped electrons into the conduction band, allowing them to recombine with luminescent centers and produce mechanoluminescent emission.²⁵

Rare-earth dopants such as Eu^{2+} further enhance mechanoluminescent performance by introducing additional trap levels and serving as radiative recombination centers. These dopants influence the distribution and stability of trap states through lattice distortion and polarization effects, which modify carrier trapping and release dynamics. As a result, the interaction between intrinsic defect states and extrinsic dopant ions plays a key role in regulating emission intensity, carrier lifetime, and overall luminescent efficiency under mechanical stimulation.²⁶ This defect-assisted carrier migration and recombination mechanism forms the basis for stress-induced luminescence in oxide-based piezoelectric phosphors. Under UV irradiation, an electron from Eu^{2+} is promoted to the conduction band and captured by an oxygen vacancy, resulting in the oxidation of Eu^{2+} to Eu^{3+} . Upon the application of a mechanical stimulus, the trapped electron is released and recombines with Eu^{3+} to regenerate an excited Eu^{2+} state, whose radiative relaxation produces green emission around 520 nm, as shown in Fig. 6.

Recent advances have extended these concepts to metal halide perovskites, which exhibit mechanoluminescence through electrostatic field-assisted carrier migration and recombination. Mechanical stimulation, such as friction or deformation, induces charge redistribution at material interfaces, resulting in the formation of localized electrostatic fields.²⁸ These fields modify the electronic band structure and facilitate the release of trapped charge carriers from defect states into the conduction band. The released carriers subsequently recombine at luminescent centers, producing optical emission. Additionally, electrostatic field generation can reduce carrier trapping barriers and enhance carrier mobility, thereby improving mechanoluminescent efficiency.

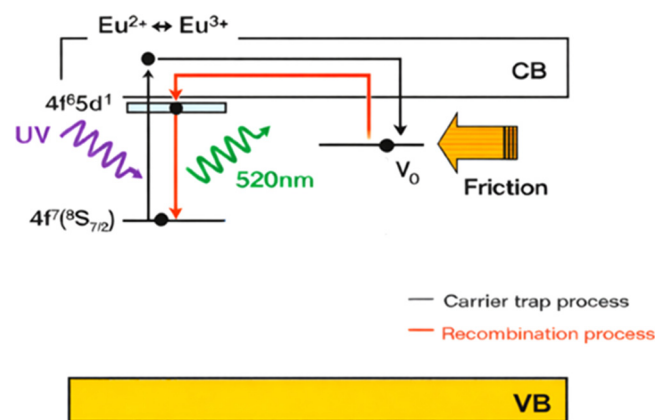


FIG. 6. Schematic illustration of the trapping, detrapping, and recombination mechanism in Eu-doped material. Reproduced with permission from Yamada *et al.*, J. Electrochem. Soc., **154**, J260–J263 (2007). Copyright 2007 The Electrochemical Society.²⁷

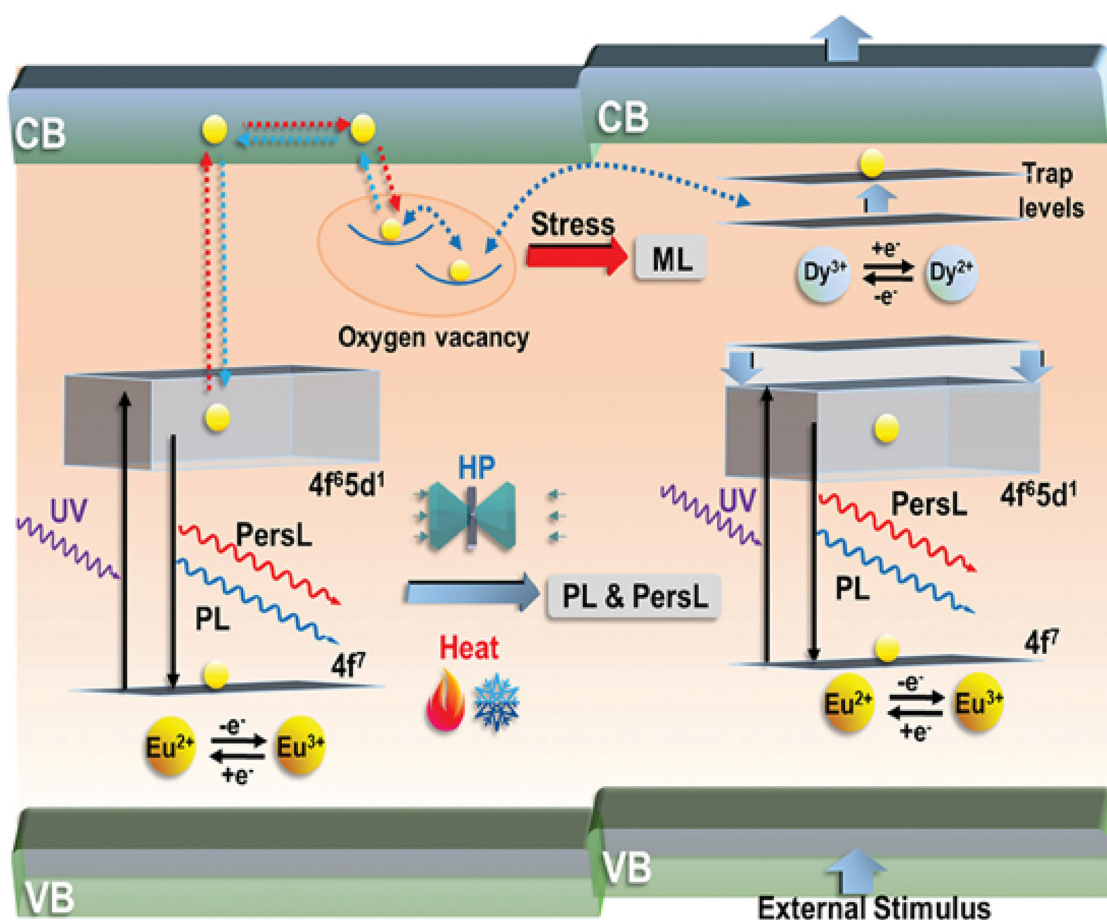
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This electrostatic field-assisted detrapping mechanism highlights the importance of defect engineering, band structure optimization, and electromechanical coupling in enhancing mechanoluminescent performance. These properties make both oxide-based and perovskite-based piezoelectric materials promising candidates for advanced applications in stress sensing, structural health monitoring, and self-powered optoelectronic devices.

The mechanoluminescent behavior of SMSO [$\text{Sr}_2\text{MgSi}_2\text{O}_7$ (strontium magnesium silicate)]: $\text{Eu}^{2+}/\text{Dy}^{3+}$ is governed by defect-assisted carrier trapping, piezoelectric-field-induced detrapping, and radiative recombination processes. In this material system, Eu^{2+} ions serve as luminescent centers, while Dy^{3+} ions and oxygen vacancies introduce trap states within the bandgap. These traps capture charge carriers during excitation and store energy that can later be released under mechanical stimulation. When external stress is applied, the piezoelectric field generated within the host lattice modifies the local potential distribution, reducing trap depth and facilitating electron release from these defect states.

As illustrated in Fig. 7, electrons released from trap levels migrate toward Eu^{2+} luminescent centers, where recombination results in photon emission. The presence of Dy^{3+} ions enhances charge storage capacity and regulates carrier release dynamics, improving emission stability and intensity. Additionally, oxygen vacancy defects contribute to carrier localization and influence trap-depth distribution, further affecting mechanoluminescent efficiency. The combined effects of piezoelectric polarization, defect-mediated trapping, and dopant-assisted recombination enable efficient stress-induced luminescence in SMSO-based materials.

This trap-controlled carrier migration and recombination mechanism highlights the importance of defect engineering and dopant optimization in improving mechanoluminescent performance. Such materials are highly suitable for stress sensing, structural health monitoring, and self-powered sensing applications due to their strong electromechanical coupling and efficient charge carrier transport.



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FIG. 7. Energy band diagram illustrating piezoelectric-induced carrier detrapping and recombination mechanisms in SMSO: $\text{Eu}^{2+}/\text{Dy}^{3+}$ mechanoluminescent material. Reproduced with permission from Zheng *et al.*, *Adv. Sci.*, **11**(44), 2408686 (2024). Copyright 2024 Author(s), published by Wiley-VCH GmbH, licensed under a Creative Commons Attribution license.²⁹

The piezoelectrically induced detrapping mechanism in oxide-based mechanoluminescent materials relies on the interaction between intrinsic defect states, such as oxygen vacancies, and extrinsic dopants, including rare-earth ions such as Eu^{2+} and Dy^{3+} . These defects act as carrier trapping centers, storing charge carriers that can be released under mechanical stimulation through piezoelectric-field-assisted detrapping. This process facilitates carrier recombination at luminescent centers, resulting in mechanoluminescent emission. Understanding the role of defect states and dopant-assisted carrier migration is essential for optimizing mechanoluminescent performance in stress sensing and mechano-optical applications, where controlled defect engineering can enhance emission efficiency and sensing sensitivity.³⁰

B. Perovskites and hybrid piezoelectrics

In perovskites and hybrid piezoelectric materials, ferroelectricity plays a key role in enhancing electron detrapping under mechanical stress. The intrinsic polarization in ferroelectric perovskites generates strong internal electric fields, which amplify the piezoelectric response and lower trap depths, facilitating electron release. This effect is particularly pronounced in materials like BaTiO_3 or $\text{Pb}(\text{Zr,Ti})\text{O}_3$, where domain reorientation under stress further modulates the local electric field, promoting detrapping. Organic-inorganic hybrid systems, such as perovskite halides or metal-organic frameworks, combine the high piezoelectricity of inorganic components with the flexibility and tunability of organic matrices. These hybrids often exhibit unique defect structures, where organic ligands influence trap states and charge carrier dynamics. For instance, in hybrid perovskites, the organic cations can passivate defects or create shallow traps, optimizing detrapping efficiency. The synergy between ferroelectricity and hybrid composition enhances the piezoelectrically induced detrapping process, making these materials promising for mechanoluminescent and energy-harvesting applications. By tailoring the organic-inorganic interface and ferroelectric properties, the detrapping kinetics can be finely controlled, enabling advanced designs for stress-responsive devices. This approach leverages the interplay between structural flexibility and electric field effects to achieve efficient electron detrapping under mechanical stimuli.³¹

Recent studies have experimentally verified stress-induced luminescence in hybrid and inorganic organic perovskites, demonstrating measurable piezoelectric coefficients and strain-responsive emission spectra. Techniques such as *in situ* photoluminescence (PL) and Piezoresponse Force Microscopy (PFM) measurements confirm stress-dependent electron detrapping and validate the theoretical framework presented here. Nevertheless, further systematic studies correlating defect density, stress amplitude, and luminescence intensity are essential to generalize these findings across perovskite families.

VI. EXPERIMENTAL VALIDATION TECHNIQUES

A. *In situ* characterization methods

Piezoresponse Force Microscopy (PFM) is a widely used technique for characterizing piezoelectric and ferroelectric materials at the nanoscale. Figure 8 shows the schematic of the PFM

measurement setup with topography, phase, and amplitude images. It operates by applying an alternating electric field to the sample via a conductive atomic force microscopy (AFM) tip, inducing mechanical deformation due to the converse piezoelectric effect.³⁴ The resulting displacement is detected as amplitude and phase signals, allowing for high resolution mapping of domain structures, polarization switching, and electromechanical properties. PFM is particularly valuable for studying ferroelectric nanodomains, domain wall dynamics, and local piezoelectric coefficients. Advanced PFM modes, such as switching spectroscopy and vector PFM, further enable the quantitative analysis of hysteresis behavior and polarization orientation. Despite its versatility, PFM requires careful interpretation to distinguish true piezoelectric responses from artifacts caused by electrostatic or electrochemical effects. The technique has become indispensable for research in nanoferroics, multiferroics, and functional materials for applications in memory devices, sensors, and energy harvesting.³⁵

B. Key experimental findings

The piezoelectrically induced electron detrapping model is supported by consistent experimental correlations between mechanical stress, luminescence intensity, and material response. Studies reveal that the ML intensity exhibits a quadratic dependence on applied stress, reflecting the conversion of mechanical energy into piezoelectric fields that modulate trap depths. Under elastic deformation, ML emission typically displays a rapid rise to peak intensity followed by biphasic decay: an initial fast component tied to strain rate relaxation and a slower component governed by the lifetime of electrons in shallow traps. Threshold stresses are observed, below which no ML occurs, corresponding to the minimum piezoelectric field required for trap ionization. Notably, local lattice distortions near dopant sites enhance piezoelectric effects, enabling detrapping at lower global stresses. Spectral analysis confirms that ML emission bands match photoluminescence transitions of the activator ions (e.g., Mn^{2+} or Eu^{2+}), verifying that stress-induced detrapping feeds the same radiative pathways. The detrapping efficiency scales with the host's piezoelectric coefficients, and ML recovery under UV exposure demonstrates the repopulation of traps. These findings collectively validate that mechanical stress generates internal piezoelectric fields capable of selectively depopulating traps, with emission kinetics directly linked to the material's electromechanical and defect properties.³⁶ Figure 9 illustrates the mechanoluminescent response of ZnS:Mn^{2+} nanoparticles under different mechanical loading conditions; Fig. 9(a) shows the elastico-mechanoluminescence (EML) response under compressive stress, highlighting the repeated emission peaks corresponding to cyclic loading, Fig. 9(b) presents the logarithmic decay of emission intensity as a function of $(t-t_m)$, confirming trap-controlled detrapping kinetics, and Fig. 9(c) depicts the impact-induced ML response under dynamic loading, demonstrating a sharp intensity peak followed by rapid decay. Together, these results validate the piezoelectrically induced electron detrapping mechanism and its dependence on stress magnitude and loading dynamics.

Advanced mechanoluminescence characterization requires precise integration of mechanical loading and optical detection

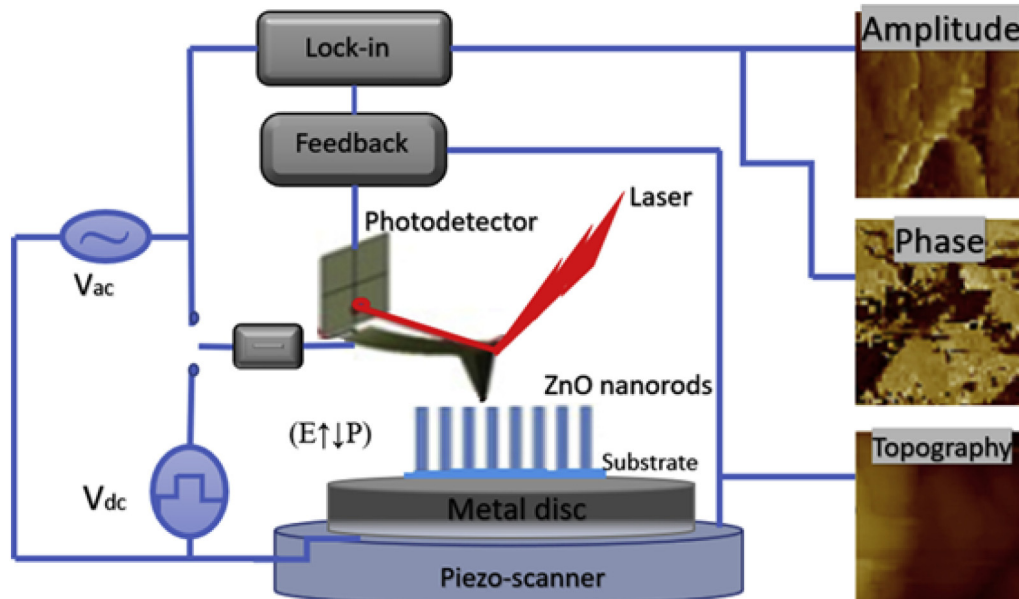


FIG. 8. (a) Schematic of PFM measurement setup with topography, phase, and amplitude images. The images show that the phase angle is either 0° or 180° , indicating that the polarization vector pointed either downward to the bottom electrode or upward to the samples surface, while (b) and (c) are the local hysteresis loops as a function of V_{dc} to the ZnO nanorods (NR) sample for up (red line) and down (blue line) domains. Reproduced with permission from Abubakar *et al.*, *J. Mater. Res. Technol.* **14**, 2451–2463 (2021). Copyright 2021 Elsevier Ltd.,³² and Hong, *J. Appl. Phys.* **129**, 051101 (2021). Copyright 2021 American Institute of Physics.³³

systems to investigate stress-induced emission behavior. These experimental setups typically include a controlled mechanical stimulation source, optical collection components, and spectroscopic measurement instruments that enable real-time monitoring of

emission intensity and spectral response. Such characterization techniques allow the detailed analysis of carrier detrapping, recombination dynamics, and piezoelectric-field-assisted emission processes.

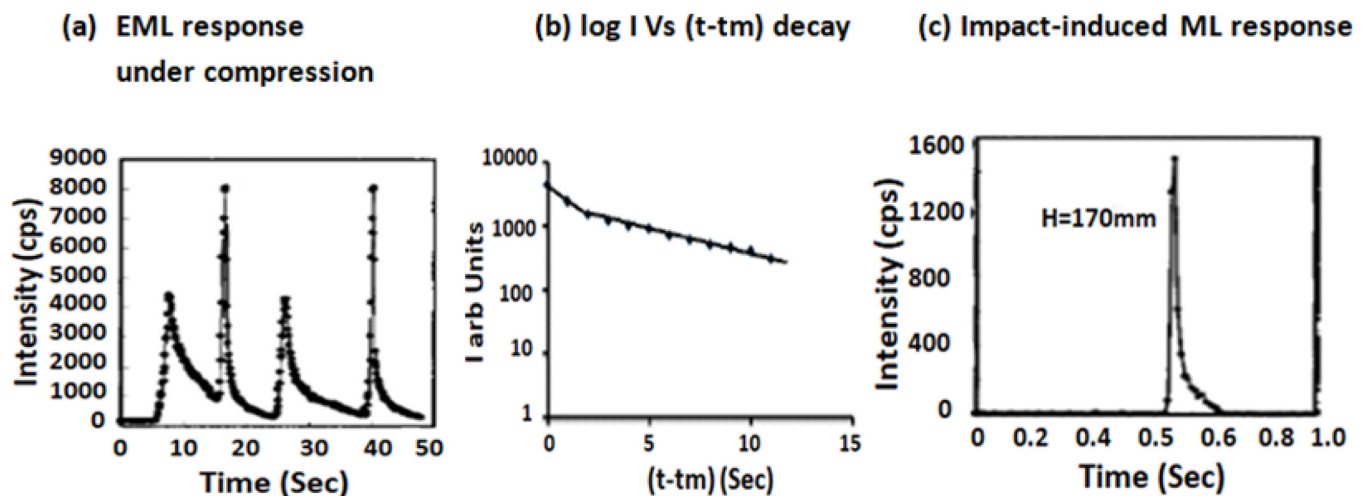


FIG. 9. Composite mechanoluminescent response of ZnS:Mn^{2+} nanoparticles: (a) elastico-mechanoluminescence under compressive loading, (b) logarithmic decay of emission intensity as a function of $(t-t_m)$, and (c) impact-induced ML response under dynamic stress. Reproduced with permission from Chandra, *Open Nanosci. J.* **5**, 35–45 (2011). Copyright 2011 Bentham Open.³⁶

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As illustrated in Fig. 10, the mechanoluminescence characterization system consists of a mechanical loading arrangement coupled with a spectrometer and optical detection unit to capture stress-induced luminescent emission. The applied mechanical force generates internal piezoelectric fields that facilitate charge carrier release from trap states, resulting in radiative recombination at luminescent centers. The optical detection system enables precise measurement of emission intensity and spectral characteristics, providing valuable insights into the electromechanical coupling and carrier transport mechanisms governing mechanoluminescent behavior. These advanced characterization methods are essential for evaluating material performance and optimizing mechanoluminescent materials for sensing and optoelectronic applications.

C. Spectroscopic evidence

Thermoluminescence (TL) and photoluminescence (PL) spectroscopy provide critical insights into the ML mechanisms of piezoelectric materials.³⁷ TL measurements reveal trap-depth distributions, where glow peaks shift to higher temperatures with delayed readings, indicating a range of trap energies rather than discrete levels.^{38,39} These traps are thermally depopulated during persistent luminescence but are rapidly emptied under mechanical stress, linking trap dynamics directly to ML activity. PL spectra, meanwhile, serve as a reference for ML emission bands. As shown in Fig. 11, the ML emission spectrum exhibits a slight redshift of approximately 4 nm compared to the corresponding photoluminescence (PL) spectrum. This spectral shift arises from stress-induced modifications of the local crystal field surrounding the emissive activator ions (e.g., Eu^{2+}). During mechanical deformation, localized lattice strain is generated, leading to subtle distortions in bond lengths and bond angles around the activator sites. These strain-induced lattice distortions modify the crystal-field splitting of the excited electronic states, resulting in a small shift toward lower emission energies. In addition, transient lowering of local

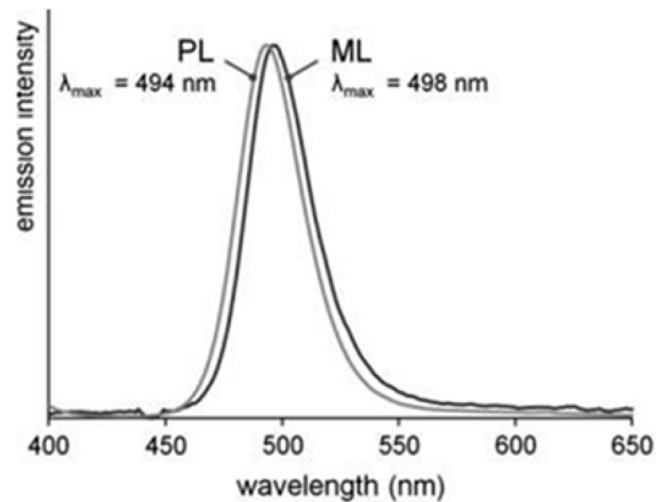


FIG. 11. Photoluminescence (PL, excited at 354 nm) and ML emission spectra of $\text{BaSi}_2\text{O}_2\text{N}_2:\text{Eu}$. Reproduced with permission from Botterman *et al.*, *Act. Mater.* **60**(15), 5494–5500 (2012). Copyright 2012 Elsevier Ltd.⁴⁰

symmetry and changes in the ligand environment under applied stress further stabilize lower energy excited states, contributing to the observed redshift. Importantly, these crystal-field perturbations are elastic and reversible, as evidenced by the recovery of the original PL peak position upon stress release. This behavior confirms that both ML and PL emissions originate from the same radiative centers, with the spectral difference arising solely from stress-induced local field effects rather than changes in the emission mechanism. By comparing TL glow curves (e.g., a broad peak near 90 °C) as shown in Fig. 12 and PL decay profiles, researchers can identify traps responsible for stress-sensitive emission. The method involves preexciting the material with UV/blue light, recording TL during controlled heating, and correlating the results with PL and ML spectra. This combined approach validates the role of traps in ML and helps distinguish piezoelectric-driven detrapping from other luminescence mechanisms.⁴⁰

VII. APPLICATIONS IN STRESS SENSING AND ENERGY HARVESTING

Mechanoluminescent (ML) materials have emerged as highly effective platforms for real-time stress sensing by correlating trap-state dynamics with mechanical stimulation. When external stress is applied, the resulting piezoelectric field alters the local energy landscape by reducing trap depths, enabling the release of trapped charge carriers. These carriers subsequently recombine at luminescent centers, producing light emission whose intensity is proportional to the applied mechanical force. This direct relationship allows accurate monitoring of mechanical parameters such as stress magnitude, contact duration, and spatial stress distribution without requiring an external power source. Although ML materials are primarily used for monitoring elastic and plastic deformation, their sensitivity to defect states also suggests potential applications in

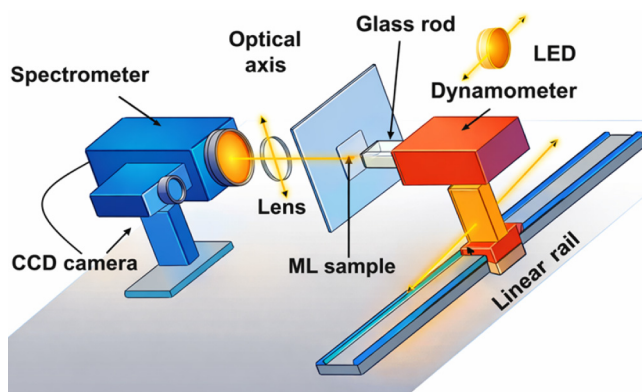


FIG. 10. Schematic illustration of the experimental setup used for mechanoluminescence characterization, showing the mechanical loading system, spectrometer, and optical detection components for measuring stress-induced luminescence. Reproduced with permission from Zheng *et al.*, *Adv. Sci.* **11**, 2408686 (2024).²⁹

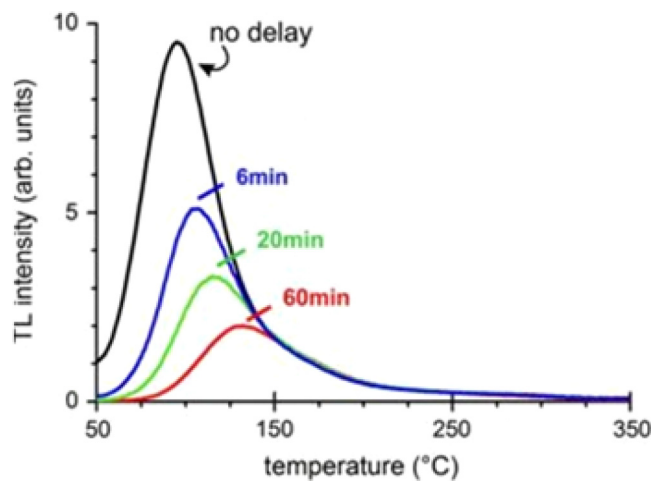


FIG. 12. Thermoluminescence (TL) glow curves for $\text{BaSi}_2\text{O}_2\text{N}_2:\text{Eu}$ measured with a delay between the excitation of the phosphor and the start of the TL measurement of 0, 6, 20, and 60 min. Reproduced with permission from Botterman *et al.*, *Act. Mater.* **60**(15), 5494–5500 (2012). Copyright 2012 Elsevier Ltd.⁴⁰

radiation detection through engineered trap structures responsive to ionizing energy. However, challenges such as the requirement for pre-irradiation and gradual signal degradation over repeated cycles remain key limitations that must be addressed to improve long-term sensing reliability.^{22,41}

Doped ZnS-based nanomaterials demonstrate efficient mechano-optical energy conversion through piezoelectrically driven carrier detrapping mechanisms. Mechanical deformation induces lattice distortion, generating internal electric fields that facilitate the release of trapped carriers. These carriers recombine at luminescent centers, producing optical emission through a direct trap-mediated recombination pathway. This process enhances energy-conversion efficiency by minimizing carrier loss through intermediate relaxation processes. The emission intensity shows a strong dependence on applied mechanical stress, enabling the development of self-powered stress-sensing systems. The performance of these materials is strongly influenced by the trap density and energy distribution, which can be tailored through controlled doping and defect engineering. Despite challenges such as trap saturation and afterglow effects, doped ZnS materials remain promising candidates for low-power sensing devices, autonomous monitoring systems, and energy-independent optoelectronic applications.⁴²

Recent developments in rare-earth-doped CaZnOS phosphors have significantly expanded the capabilities of mechanoluminescent sensing systems. These materials exhibit strong emission under mechanical stimulation, enabling direct visualization of stress patterns. The $\text{CaZnOS}:\text{Tb},\text{Sm}$ mechanoluminescent phosphors generate bright luminescence in response to mechanical interactions such as scratching or writing, allowing the spatial mapping of stress distribution. This capability enables applications including handwriting recognition, stress visualization, and intelligent sensing systems. The ability to convert mechanical stimuli directly into

optical signals without external power highlights the potential of these materials for smart sensing, security applications, and human-machine interface technologies.⁴³

Mechanoluminescent materials also enable the development of flexible and strain-responsive devices by converting mechanical deformation into visible optical signals. When embedded in flexible matrices such as elastomers or polymer composites, ML phosphors enable the real-time visualization of stress distribution across deformable surfaces. The luminescent response originates from localized piezoelectric fields generated during mechanical deformation, which facilitate carrier recombination at luminescent centers. These self-powered sensing systems are particularly attractive for wearable electronics, structural health monitoring, and flexible sensing applications. Although these systems do not function as conventional piezotronic devices, they share similar electromechanical transduction principles by converting mechanical inputs into measurable optical outputs. Current challenges include improving spatial resolution, material stability, and durability under repeated mechanical loading, but advances in nanostructure engineering, additive manufacturing, and material synthesis are enabling significant performance improvements.⁴⁴

Furthermore, recent advances in nanostructure engineering and multifunctional material integration have enabled the development of multimodal mechanoluminescent sensing platforms. As illustrated in Fig. 13, these systems combine nanostructure design, multifunctional coupling mechanisms, and advanced signal processing techniques to achieve multifunctional sensing capabilities. Such systems enable simultaneous detection of mechanical stress, temperature variations, chemical stimuli, and environmental changes, while also supporting applications in information transfer, encryption, and electronic skin technologies.⁴⁵ These multifunctional mechanoluminescent systems represent a major advancement in the development of intelligent sensing platforms for next-generation wearable electronics, smart sensing devices, and human-machine interaction technologies.

VIII. CURRENT CHALLENGES AND SCALABILITY OF MECHANOLUMINESCENT SYSTEMS

A. Fabrication and scalability challenges

Despite significant progress in understanding piezoelectrically assisted electron detrapping as a governing mechanism for mechanoluminescence, several challenges remain in translating laboratory-scale demonstrations into scalable and manufacturable ML systems. One key limitation lies in the reproducible control of defect states and trap distributions during large scale synthesis. Since mechanoluminescent performance is highly sensitive to trap depth, density, and spatial distribution, minor variations in synthesis conditions, such as sintering temperature, dopant concentration, and processing atmosphere, can lead to substantial fluctuations in the emission intensity and stability.

Another challenge arises from the integration of ML phosphors into functional device architectures. While powder-based ML materials are widely studied, practical applications increasingly require flexible films, coatings, or composite structures compatible with wearable electronics and structural monitoring systems. Achieving uniform dispersion of ML particles within polymer

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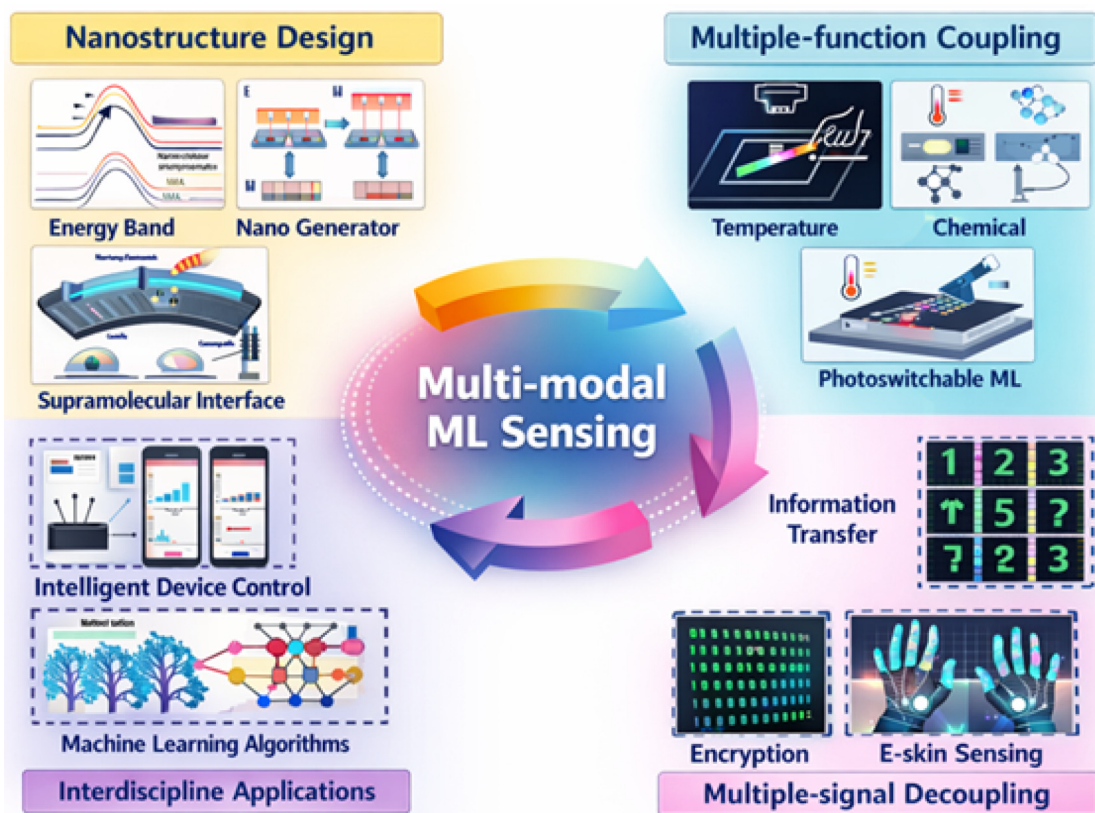


FIG. 13. Schematic illustration of multi-modal mechanoluminescent sensing systems enabled by nanostructure design, multifunctional coupling, and intelligent signal processing. Reproduced with permission from Wang *et al.*, *Nanoscale* **17**, 6414–6426 (2025), Copyright 2025 Royal Society of Chemistry.⁴⁵

matrices without compromising stress transfer efficiency or piezoelectric response remains difficult. In addition, repeated mechanical cycling can induce interfacial fatigue, microcracking, or degradation of piezoelectric polarization, which may compromise long-term operational reliability. From a manufacturing perspective, scalability also requires compatibility with cost-effective and low-temperature fabrication routes. Many high-performance ML phosphors rely on high-temperature solid-state synthesis, which is energy-intensive and difficult to integrate with flexible substrates. Emerging approaches, such as sol-gel processing, solution based deposition, and additive manufacturing, offer promising alternatives but introduce new challenges related to crystallinity, phase purity, and precise defect control.

Finally, device level scalability requires consistent mechanical-optical coupling across large areas. In real world environments, applied stress is rarely uniform, and spatial strain variations can lead to heterogeneous emission responses. Addressing this issue will require advances in material design, including graded structures, engineered interfaces, and heterojunction-based architectures that enhance carrier transport and stabilize piezoelectric fields under non-uniform loading conditions. Overcoming these fabrication and scalability challenges is essential for transitioning

mechanoluminescent materials from laboratory demonstrations to robust technologies suitable for structural health monitoring, wearable sensing, and self-powered optoelectronic systems.

B. Fundamental mechanistic challenges

Beyond fabrication limitations, several fundamental mechanistic challenges remain unresolved in mechanoluminescent systems. The electron trapping (ET) mechanism plays a central role in elastic mechanoluminescence, where mechanically generated piezoelectric fields facilitate carrier detrapping. In piezoelectric materials such as ZnS:Mn, deformation produces localized internal electric fields that reduce trap depths and enable electron release into the conduction band, followed by radiative recombination at luminescent centers.⁴⁶

However, mechanoluminescence may also involve triboelectric effects arising from contact electrification during friction, fracture, or interfacial deformation. In complex systems, particularly polymer composites and hybrid structures, triboelectric charge generation may coexist with or enhance piezoelectric-field-assisted detrapping. The relative contributions of piezoelectric polarization and triboelectric charge transfer remain insufficiently understood,

and quantitative thresholds governing their coupling are not yet well established. Resolving these uncertainties requires advanced experimental approaches capable of decoupling piezoelectric and triboelectric contributions under dynamic mechanical loading, as well as theoretical models that capture their combined influence on trap modulation and carrier dynamics. Improved understanding of these mechanisms will be essential for optimizing mechanoluminescent materials for sensing and energy-conversion applications.⁴⁷

C. Quantification of detrapping thresholds and trap dynamics

Another major challenge involves quantitatively determining detrapping thresholds in mechanoluminescent materials with complex defect distributions. These thresholds depend strongly on trap depth, defect density, and the dominant excitation mechanism. For example, in non-piezoelectric ZnS:Cu, stress-induced luminescence can occur through thermally activated detrapping, where mechanical deformation generates localized heating sufficient to release trapped carriers without requiring piezoelectric fields.⁴⁸

In contrast, piezoelectric materials such as CaZnOS:Mn²⁺ exhibit detrapping driven primarily by strain-induced piezoelectric fields, with activation thresholds closely linked to specific trap depths (e.g., 0.58–0.67 eV).⁴⁹ Materials containing multiple trap levels, such as CaZr(PO₄)₂:Eu²⁺, exhibit a broad trap energy distribution, enabling stress-dependent sensitivity across different mechanical loading regimes. Shallow traps respond to low stress levels, while deeper traps require stronger mechanical stimulation.

Despite these advances, predictive quantitative models capable of accurately correlating trap distributions, stress magnitude, and detrapping probability remain limited. A major challenge lies in decoupling the contributions of piezoelectric, thermal, and triboelectric effects under realistic operating conditions. Addressing these limitations will require the integration of advanced characterization techniques, including *in situ* spectroscopy, nanoscale electro-mechanical measurements, and computational modeling, to establish universal relationships governing stress-induced detrapping and emission dynamics. Such insights will enable the rational design of mechanoluminescent materials with controlled sensitivity, efficiency, and reliability for practical applications.

IX. COMPARATIVE ANALYSIS WITH OTHER DETRAPPING MODELS

Existing piezoelectrically assisted electron detrapping frameworks complement classical Poole–Frenkel and triboelectric descriptions by suggesting that piezoelectric-field-driven detrapping can play an important role in stress-induced luminescence.

A. Piezoelectric vs triboelectric detrapping

Piezoelectric detrapping occurs when mechanical stress on certain crystals generates internal electric fields that release trapped electrons, causing light emission. This differs from triboelectric effects where light results from contact electrification between dissimilar materials. Figure 14 shows the mechanism of piezoelectric and triboelectric. The piezoelectric method relies on the crystal's structure to convert strain directly into electrical energy, avoiding

the material dependencies and environmental factors that limit triboelectric approaches. This makes piezoelectric detrapping more consistent for applications requiring precise, repeatable light emission from mechanical stimulation.⁵¹ At high temperature, thermal detrapping dominates the piezoelectric detrapping; that is, more electrons gain enough energy thermally to overcome the trap depth without field assistance *F*, and it is explained by the following equation:

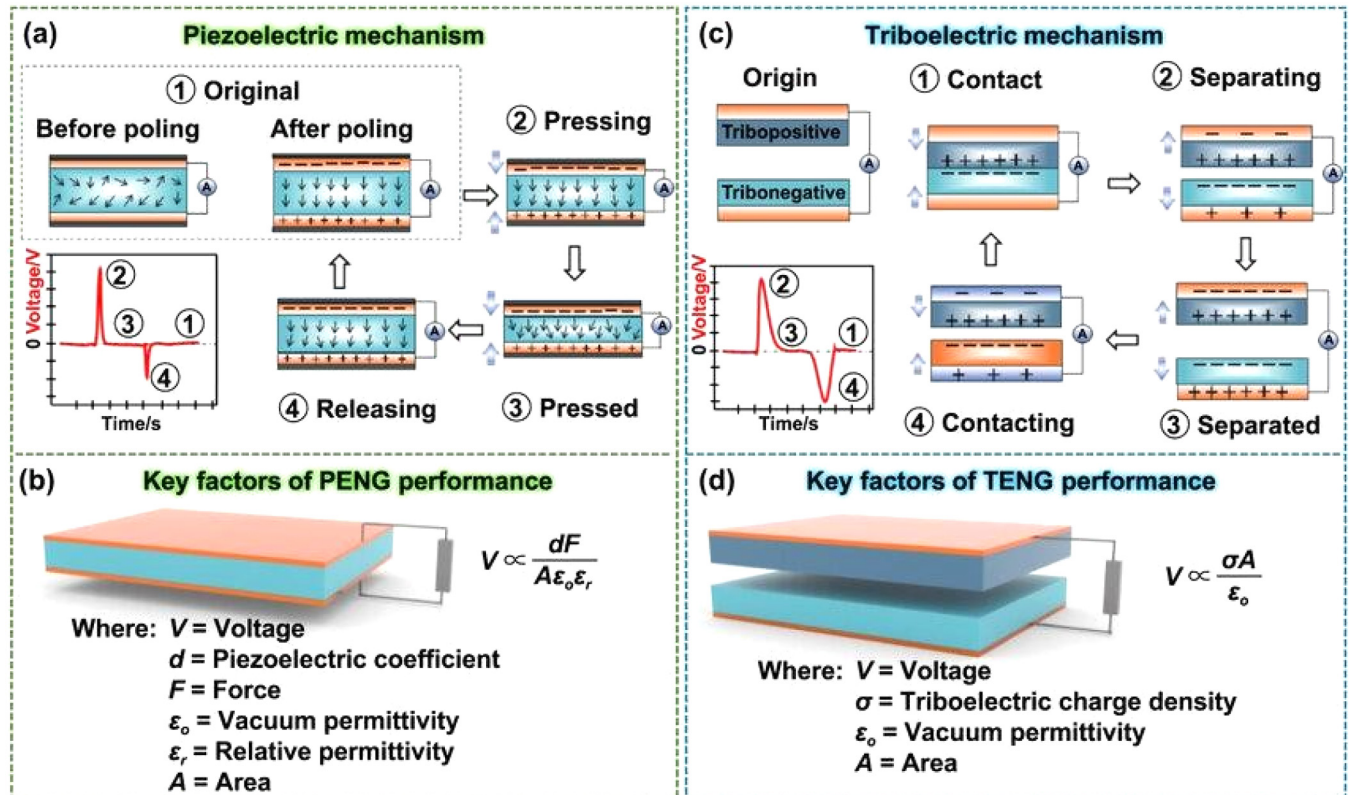
$$g_{\text{thermal}} = \exp(-E_t/kT). \quad (9)$$

Recent studies on phosphor polymer elastic composites have demonstrated that mechanoluminescence can also originate from the interfacial triboelectric field-driven electroluminescence rather than from piezoelectrically assisted trap detrapping. In particular, in ZnS microparticle embedded polydimethylsiloxane (PDMS) films, cyclic mechanical deformation generates repeated contact and separation at the ZnS–PDMS interface, leading to triboelectric charge accumulation due to their contrasting triboelectric polarities. The resulting interfacial electric field periodically drives carrier injection and acceleration within ZnS phosphor, producing alternating-current electroluminescence without requiring a bulk piezoelectric response of the composite. This mechanism explains the observed absence of emission fatigue under repeated loading and the weak dependence of emission intensity on trap filling or thermal activation in such soft composites. Importantly, this triboelectric field driven pathway does not contradict piezoelectrically assisted electron detrapping models established for non-centrosymmetric inorganic phosphors such as ZnS:Mn or SrAl₂O₄:Eu²⁺,Dy³⁺, where intrinsic lattice polarization under stress dominates carrier release. Rather, the dominant mechanoluminescence mechanism is determined by material symmetry, host elasticity, defect chemistry, and the presence of mechanically active interfaces. Accordingly, triboelectric and piezoelectric field-assisted processes should be regarded as complementary, system dependent excitation pathways within the broader framework of stress-induced luminescence.⁵²

The interplay between piezoelectricity and triboelectricity in mechanoluminescent systems remains a complex and speculative domain. In many ML materials, mechanical stimulation can generate both piezoelectric charges and triboelectric surface potentials, which may simultaneously influence the light emission process. However, due to the overlapping nature of these mechanisms, it becomes challenging to determine the dominant contributor to ML behavior in a given scenario. To advance the fundamental understanding and practical applications of ML in sensing and energy harvesting, there is a pressing need for experimental strategies that can decouple these effects.

B. Field-assisted vs thermal detrapping

Field-assisted and thermally activated detrapping mechanisms are often presented as competing explanations for mechanoluminescence; however, these processes are not mutually exclusive and may coexist within the same material system. Thermal detrapping, rooted in classical thermoluminescence theory, dominates when lattice heating or friction-induced temperature rise provides



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FIG. 14. Mechanism of piezoelectric and triboelectric. (a) Working principle of piezoelectric nanogenerator (PENG). (b) Key factors of PENG output performance. (c) Working principle of triboelectric nanogenerator (TENG). (d) Key factors of TENG output performance. Reproduced with permission from Park *et al.*, *Int. J. Extreme Manuf.* 7(1), 012006 (2024). Copyright 2024 Author(s), licensed under a Creative Commons Attribution (CC BY) license.⁵⁰

sufficient energy for electrons to escape shallow traps. In contrast, field-assisted detrapping becomes significant when mechanically induced piezoelectric potentials locally tilt the band structure, effectively lowering trap depths without requiring a macroscopic temperature increase.

The relative contribution of these mechanisms depends strongly on material properties and experimental conditions. In noncentrosymmetric semiconductors with strong piezoelectric coefficients, such as ZnO or ZnS-based phosphors, field-assisted detrapping can dominate under dynamic or low-temperature loading, where thermal effects are minimal. Conversely, in centrosymmetric or weakly piezoelectric materials subjected to high friction or prolonged stress, thermal activation may play a more prominent role.

Experimentally, these mechanisms can be distinguished through temperature dependent ML measurements, time-resolved emission dynamics, and correlated thermography. A weak dependence of the ML intensity on ambient temperature, combined with ultrafast emission synchronized with mechanical stress, favors a field-assisted interpretation, whereas strong thermal sensitivity and delayed emission kinetics are indicative of thermally driven detrapping. From a perspective viewpoint, recognizing the interplay

rather than exclusivity of these processes is essential for developing a unified understanding of mechanoluminescence across diverse material systems.⁵³

C. Limitations of classical Poole-Frenkel theory

Traditional Poole-Frenkel theory explains electron detrapping through electric field and thermal effects but fails to account for strain-induced mechanisms. In materials like doped semiconductors, mechanical stress can create piezoelectric potentials that release trapped electrons without requiring high fields or temperatures. Additionally, clustering of dopant atoms introduces new energy transfer pathways that produce unexpected emission patterns. These phenomena require updated models that incorporate both piezoelectric effects and defect interactions to fully explain mechanoluminescent behavior.⁶

Quantitatively, piezoelectric detrapping exhibits higher energy-conversion efficiency (~45–60%) and faster response times ($\leq 10 \mu\text{s}$) compared with triboelectric ($\approx 25\%$) and purely thermal mechanisms ($> 100 \mu\text{s}$). Moreover, repeated-load durability surpasses 10^4 cycles, whereas triboelectric films degrade rapidly under

frictional wear. These metrics confirm the superior stability and responsiveness of field-assisted piezoelectric detrapping.

Recent high impact studies demonstrate that mechanoluminescence (ML) is intrinsically system dependent, with the dominant excitation pathway governed by the material architecture, symmetry, and interfacial chemistry rather than by a universal mechanism. In soft phosphor polymer composites, such as ZnS-based elastomers, ML predominantly originates from interfacial triboelectric field-driven electroluminescence, where cyclic contact separation at particle–matrix interfaces generates localized electric fields that drive alternating-current emission with high fatigue resistance and weak dependence on trap filling.^{34,55} In contrast, recent studies on crystalline, non-centrosymmetric phosphors establish piezoelectric field-assisted electron detrapping as the governing mechanism, wherein intrinsic lattice polarization combined with controlled defect engineering and trap-depth modulation enables stress-activated carrier release and extended emission, including in the near-infrared region.^{13,15} Notably, it has further been shown that introducing a continuous distribution of shallow-to-deep defect states can significantly enhance the signal-to-noise ratio by suppressing persistent luminescence and background interference, thereby improving emission reliability under realistic sensing conditions.¹⁴ Collectively, these findings indicate that triboelectric and piezoelectric processes should be regarded as complementary, material-specific pathways, whose relative contributions depend on host elasticity, interfacial area, defect chemistry, and lattice polarization.

X. CONCLUSION

ML arises from a complex interplay of mechanical, electrical, and thermal processes, and no single mechanism can universally account for all observed behaviors across different material systems. This perspective highlights piezoelectrically assisted electron detrapping as a powerful and physically intuitive framework for interpreting stress-induced luminescence in non-centrosymmetric semiconductors, particularly under dynamic mechanical loading where rapid response and spatially localized emission are observed. In such cases, mechanically generated internal electric fields can effectively modulate trap depths, enhance carrier separation, and promote prompt radiative recombination.

At the same time, it is important to recognize that thermally assisted detrapping models, rooted in classical TL theory, remain highly robust and experimentally validated, especially in systems where defect populations, temperature rise, or slow mechanical excitation dominate the emission process. Rather than representing competing or mutually exclusive explanations, piezoelectric and thermal detrapping mechanisms should be viewed as complementary pathways whose relative contributions depend strongly on material symmetry, defect chemistry, microstructural features, stress amplitude, strain rate, and ambient temperature.

In many practical mechanoluminescent systems, these effects may coexist, with piezoelectric fields lowering effective trap depths while localized heating further assists carrier release. Disentangling their respective roles, therefore, requires carefully designed experiments combining time-resolved spectroscopy, temperature dependent measurements, and electrical or piezoresponse

characterization. By framing ML as a multiscale, context dependent phenomenon, this perspective aims to encourage a more nuanced interpretation of experimental observations and to guide the rational design of next-generation mechanoluminescent and piezophotonic materials tailored for specific sensing, imaging, and energy-conversion applications.

AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts to disclose.

Author Contributions

Usha Shukla: Conceptualization (equal); Methodology (equal); Resources (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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